



Identification of new potential antigen recognized by $\gamma\delta$ T cells in hepatocellular carcinoma

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Abstract

In recent years, the application of chimeric antigen receptor T-cell (CAR-T) therapy based on gamma delta T ($\gamma\delta$ T) cells in hepatocellular carcinoma (HCC) immunotherapy has attracted more and more attention. However, specific antigens recognized by $\gamma\delta$ T cells are rarely identified, which has become the main restriction on such therapeutic application of $\gamma\delta$ T cells. In this report, we identified a new peptide and protein antigen recognized by $\gamma\delta$ T cells in HCC using our previous established strategy. First, we investigated the diversity of the $\gamma 9/\delta 2$ T-cell immunorepertoire by sequence analyses of the expressed complementarity-determining region 3 (CDR3) in HCC patients. Then, we constructed $\gamma 9/\delta 2$ T-cell receptor (TCR)-transfected cell lines expressing significant HCC CDR3 sequence and identified a series of peptides capable of binding to $\gamma\delta$ T cells specifically. Next, we identified, further tested and verified the biological functions of these peptides and their matched protein by bioinformatics analysis. We identified that the new protein hepatocyte growth factor-like protein, also called as macrophage-stimulating protein (MSP), and peptide HP1, not only bound to HCC-predominant $\gamma\delta$ TCR but also effectively activated $\gamma\delta$ T cells isolated from HCC patients. Moreover, they could stimulate $\gamma\delta$ T cells in peripheral blood from HCC patients to produce cytokines, which contributed to inhibiting HCC and played an important role in mediating cytotoxicity to HCC cell lines. In conclusion, we identified MSP and HP1, which showed potential as candidates for antigens recognized by $\gamma\delta$ T cells in HCC.

Keywords Hepatocellular carcinoma · Antigen · $\gamma\delta$ T cells · Macrophage stimulating protein

Xueyan Xi, Yang Guo and Min Zhu contributed equally to this report.

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Introduction

Hepatocellular carcinoma (HCC) is one of the most common malignancies which is characterized by poor prognosis [1]. Although significant advances in treatments have increased survival time, finely targeted therapy is not yet available. In recent years, adoptive immunotherapy in which lymphocytes are genetically engineered to use chimeric antigen receptor T (CAR-T) cells has demonstrated promise in treating malignant diseases [2, 3]. The key to CAR-T technology is targeting tumor-specific antigens (TSAs). But, so far, only a few TSAs of HCC have been identified. Glypican-3 (GPC3), an oncofetal antigen overexpressed in HCC, is an effective target for cytotoxic T lymphocyte (CTL)-mediated tumor immunotherapy. However, this tumor response is rare, and most HCC patients are unlikely to benefit from the vaccination strategy targeting a single CTL epitope [4]. Accumulating evidence shows that tumor progression is regulated by cross-talk between tumor cells and the host immune micro-environment, including both innate and adaptive immunity

[5]. This indicates that a type of HCC therapy that coordinates the interplay between innate and adaptive immune systems is required. The liver is commonly considered an immune organ with predominantly innate immunity [6]. Up to 35% of total liver T cells in humans carry gamma delta T-cell receptors ($\gamma\delta$ TCRs) [7]. Because $\gamma\delta$ T cells have killing effects similar to those of CTLs and are not restricted by major histocompatibility complex (MHC), they provide a strong defense against tumor formation and progression in the liver [8]. The best-known antigens for $\gamma\delta$ T cells are phosphoantigens, with research progress in $\gamma\delta$ T cells, some protein antigens recognized by $\gamma\delta$ T cells are being identified continually. The most abundantly circulating $\gamma\delta$ T cells, V γ 9V δ 2T cells, can be activated by small non-peptide phosphorylated antigens, various members of the UL16-binding protein (ULBP) and MHC class I-related molecules A/B (MICA/B) in an MHC-independent manner [9–11]. However, no liver-specific antigen or epitope of activated $\gamma\delta$ T cells has been reported.

A strategy using intact cells as probes to pan target peptides in a phage display library (PDL) was developed [12, 13]. Using HepG2 cells as a probe, the researcher obtained a target peptide, HCBP1, that might be a potential candidate for drug therapy targeted to HCC [14]. In our previous study, we established a novel strategy to screen bacillus Calmette-Guerin (BCG)-specific, $\gamma\delta$ TCR-recognized peptide and protein antigens by panning a 12-mer random-peptide using transfected cell lines as probes [15]. The strategy provides a novel means of screening mycobacterial- adjuvants or vaccine candidates. Using this strategy, we found a significant complementary determinant region 3 (CDR3) sequence in pulmonary tuberculosis (TB) patients [16], and we also identified new TB protein antigens (1-deoxy-D-xylulose 5-phosphate synthase 2, DXS2) and a peptide (EP) recognized by $\gamma\delta$ TCRs in pulmonary TB patients [17].

In this report, we again applied this strategy to identify HCC antigens recognized by $\gamma\delta$ T cells. We first investigated the diversity of the γ 9/ δ 2 T-cell immunorepertoire by sequence analyses of the expressed CDR3 in HCC patients. Then, we constructed γ 9/ δ 2TCR-transfected cell lines expressing the predominant HCC CDR3 sequence and identified a series of peptides capable of binding specifically

to $\gamma\delta$ T cells. Next, we further tested and verified the biological functions of these peptides and their matched proteins as identified by bioinformatics analysis. We identified a new protein hepatocyte growth factor-like protein, also called as macrophage-stimulating protein (MSP), and peptide HP1, which not only bound to HCC-predominant $\gamma\delta$ TCRs but also effectively activated $\gamma\delta$ T cells isolated from HCC patients. Moreover, they could stimulate $\gamma\delta$ T cells from HCC patients to produce cytokines, which helped inhibit HCC and played an important role in mediating cytotoxicity to HCC cell lines.

Materials and methods

Subjects

Forty patients were enrolled in this study, all of them pathologically diagnosed with HCC at Renmin Hospital, Shiyuan City, China in 2018–2020. To control potential confounding factors, patients (mean age, 49.7; 28 men, 12 women) who received chemotherapy or radiotherapy treatment were excluded. Twenty healthy volunteer subjects (mean age, 41; 12 men, 8 women) were recruited as a control group. Healthy volunteers did not have any changes in X-rays, hepatitis B virus (HBV) history or other underlying diseases. Exclusion criteria for the control groups were medication, pregnancy and any abnormalities in liver and renal function tests. This work received approval from the Clinical Ethics Committee of Hubei University of Medicine (2019-TH-024). All individuals gave their informed consent to participate. Basic demographic information on patients and controls is listed in Table 1.

Reagents and cell lines

We used a Ph.D.-12 phage display peptide library kit (New England Biolabs, US) to screen specific peptides binding to $\gamma\delta$ TCR. We obtained J.RT3-T3.5 cells from the American Type Culture Collection (ATCC, US). Peripheral blood mononuclear cells (PBMCs) were separated from peripheral blood via density gradient centrifugation with

Table 1 Characteristics of the hepatocellular carcinoma patients and healthy controls

	Mean age Years (range)	Male	Other complication ^b	HBV positive	Treatment time > 1 year	Treatment program
Patients (<i>n</i> = 40)	49.7 (31–75)	28 (70% ^a)	12 (40%)	22 (55%)	26 (52%)	TACE ^c
Control (<i>n</i> = 20)	41 (28–55)	12 (60%)	0 (0%)	0 (0%)		

^aThe percentage of patients with the characteristic

^bOther Complication include liver cirrhosis

^cThe treatment programs: TACE: Transarterial chemoembolization

Ficoll-Hypaque (Sigma, 17144002). We sorted the $\gamma\delta$ T cells using a magnetic-activated cell sorting (MACS) system (Miltenyi, 130-050-701). PBMCs were first magnetically labeled with Biotin-Antibody Cocktail and Anti-Biotin Microbeads; then, we loaded the cell suspension onto a MACSR column, which we placed in the magnetic field of a MACS Cell Separator while retaining the magnetically labeled $\gamma\delta$ T cells within the column. Unlabeled cells were run through. After removing the column from the magnetic field, we were able to elute the magnetically retained $\gamma\delta$ T cells as the positively selected cell fraction. $\gamma\delta$ T cells were determined by flow cytometry (FCM) using FITC- $\alpha\beta$ TCRs (Biolegend, 306706) and PE- $\gamma\delta$ TCRs (Biolegend, 331209). The purity of the $\gamma\delta$ T cells was more than 90%. We stimulated the isolated $\gamma\delta$ T cells using a commercial leukocyte activation cocktail (10 $\mu\text{g/mL}$, Becton Dickinson, 550583) in the presence of RPMI 1640 medium supplemented with 12% fetal bovine serum (FBS), interleukin-2 (IL-2), penicillin, streptomycin and 5×10^{-5} M β -mercaptoethanol at 37 °C in a humidified incubator with 95% air and 5% CO₂. Peptides were synthesized by Sangon Biotech Inc., Shanghai, China; the purity of these peptides was > 95% in high-performance liquid chromatography (HPLC) analysis. We labeled half the synthesized peptides with fluorescein isothiocyanate (FITC; Sangon) at their N-terminals, so the binding ability of peptide with cells can be detected by flow cytometry directly. The HepG2 cells were collected and were subjected to 5 h of sonication at room temperature in a water bath sonicator. The extracts were then spun at 13,000 g for 5 min and the supernatants were collected. Commercial MSP protein was purchased from Abnova (H00004485-P01).

Construction of transfected cells expressing $\gamma\delta$ TCRs with predominant HCC CDR3 sequence and healthy control

Using overlapping polymerase chain reactions (PCRs), we inserted the predominant HCC CDR3 sequence into the full-length γ 9 and δ 2 chains to substitute for the original CDR3 sequence. The detailed method of construction of expressing $\gamma\delta$ TCRs were constructed as described in our previous paper [15]. Next, we evaluated the resulting cells expressing surface $\gamma\delta$ TCRs via FCM, and we isolated positive cells via flow sorting for further experiments. Thus, we developed two artificial cell lines expressing $\gamma\delta$ TCRs, with predominant HCC CDR3 sequences and healthy-control CDR3 sequences.

Flow cytometry

To bind HP1, HP2 and HP3 to cells, we incubated the transfected cells or $\gamma\delta$ T cells with FITC-conjugated peptide and control peptide at 4 °C for 30 min. Then, we measured the

percentage of $\gamma\delta$ T cells via fluorescence-activated cell sorting (FACS) using FITC- $\alpha\beta$ TCRs and PE- $\gamma\delta$ TCRs. FITC-conjugated anti-human γ 9 and PE-conjugated anti-human δ 2 monoclonal antibody (Biolegend) were used to detect the nature and purity of $\gamma\delta$ T induced by peptide and protein antigen. The specific binding of peptide and protein to $\gamma\delta$ T cells were further proved by blocking with functional $\gamma\delta$ TCR-blocking monoclonal antibody (10 ng/ μL B1.1; eBioscience, 16–9959-81). FCM was performed on a MoFlo XDP flow cytometer (Beckman Coulter, US).

Cytotoxicity assay

We used the CytoTox 96 Non-radioactive Cytotoxicity Assay (Promega, G1780) to determine cytotoxic effects of $\gamma\delta$ T cells on the HCC cell line (HepG2), which is based on the colorimetric detection of the released lactate dehydrogenase enzyme. HepG2 cells were used as target cells and seeded into 96-well plates at 1×10^4 cells per well. Next, we added the $\gamma\delta$ T cells, which we used as effector cells, to each well at effector cell: target cell ratios of 1:1, 2.5:1, and 5:1. After the effector and target cells were incubated at 37 °C for 6 h, the culture supernatants were collected to detect the lactate dehydrogenase activity according to the manufacturer's instructions.

Real-time polymerase chain reaction (RT-PCR)

We pre-coated HP1 peptide (20 μg) or HepG2 whole protein (20 μg) onto microtiter plates (Corning) and then added J.RT-T3.5 cells, HCC-specific γ 9/ δ 2TCR-transfected cells (named as test cells) and healthy control-transfected cells (named as control cells) for 24 h. Next, we extracted total ribonucleic acid (RNA) from cells and, using Moloney murine leukemia virus reverse transcriptase (MMLV-RT), transcribed it into complementary deoxyribonucleic acid (cDNA). Using the cDNA as a template, we amplified IL-2 via RT-PCR with glyceraldehyde 3-phosphate dehydrogenase (GAPDH) for reference. Gene expression analysis was performed using Bio-Rad CFX Manager software version 3.1 (Bio-Rad Laboratories, US).

Enzyme-linked immunosorbent assay (ELISA)

Using an ELISA kit (R&D Systems, US) per manufacturer's instructions, we determined that the $\gamma\delta$ T cells secreted T helper type 1 (Th₁), Th₂ and Th₁₇ cytokines upon activation by the candidate peptide and protein. We also used ELISA kit to detect the IL-2 production level of J.RT-T3.5 cells, control cells and test cells stimulated by HepG2 whole protein.

Bioinformatics analysis

The analyzed the best matches of the germline V and J gene identified by determining alignments between the sequence results and germline sequences in the IMGT database (<http://www.imgt.org>). We performed homologous analysis and sequence alignment using the Basic Local Alignment Search Tool (BLAST; National Center for Biotechnology Information <http://www.ncbi.nlm.nih.gov/blast>) to determine matched proteins of HP1, HP2 and HP3. The first 3 matched proteins were listed. We calculated a survival analysis curve for MSP protein in HCC at the OncoLnc website (<http://www.oncolnc.org/>), which link TCGA survival data to mRNA, miRNA, or lncRNA expression levels. In the oncolnc website, MSP was entered to get the Cox regression results. HCC is selected, lower percentile was set on 10 and higher percentile was set on 90, the Kaplan plot for MSP in HCC was determined. We analyzed the peptides at the Heliquet website (<http://heliquet.ipmc.cnrs.fr>). SWISS-MODEL (<http://swissmodel.expasy.org/>) was performed to predict the structure of MSP using homology modeling techniques.

Statistical analysis

All data were analyzed using SPSS version 19.0 and GraphPad 5.0 software. Independent sample *t* tests and paired sample *t* tests were used to generate and statistically analyze on the data. *P* values < 0.05 were considered statistically significant.

Results

Sequence characteristic of $\gamma 9$ and $\delta 2$ CDR3 of HCC patients

The diversity of the TCR $\gamma\delta$ CDR3 repertoire, which reflects the proliferation of specific $\gamma\delta$ T-cell clones, is one of the most important features of HCC patients. The CDR3 regions contained conserved “CA” at the N-terminus and conserved “FGXG” at the C-terminus. In contrast, the inner regions of CDR3 were composed of variable sequences. As shown in Tables S1 and S2, predominant $\gamma 9$ and $\delta 2$ CDR3 sequences in HCC were listed. Here, a common $\delta 2$ CDR3 sequence was found in PBMC of HCC patient’s groups. Our sequence analysis also revealed that most V $\delta 2$ T cell isolated from HCC patients and healthy controls also carried a hydrophobic amino acid residue (/leucine/ valine/ argine) at conserved position 97 (Table S2).

Construction of transfected cells expressing HCC-specific and non-specific $\gamma\delta$ TCRs

Based on predominant $\gamma 9$ and $\delta 2$ CDR3 sequences in HCC listed and highlighted in Tables S1 and S2, we developed two artificial cell lines expressing HCC-specific $\gamma 9/\delta 2$ TCR-transfected cells (test cells) and healthy control-transfected cells (control cells). The full-length $\gamma 9$ and $\delta 2$ chains were cloned into pREP7 and pREP9 expression vectors and co-transfected into J.RT3-T3.5 cells by electroporation. After 4 weeks of selection using G418 and hygromycin, we observed that 30% of the transfected cells expressed $\gamma\delta$ TCRs. The positive cells were further enriched by MACS and exceeded 90% (Fig. 1a). Thus, the artificial $\gamma\delta$ TCRs transfected cells were constructed successfully. They will secrete IL-2 upon stimulated by the corresponding antigens. To further check the HCC specificity of the transfected $\gamma\delta$ TCRs, we stimulated both test and control cells with protein extracts of HepG2 cells and then measured the level of secreted IL-2 by RT-PCR and ELISA. Soluble phosphoantigens, HMBPP (1 ng/mL) and protein extracts of HepG2 cells were pre-coated with 96-well plate. J.RT3-T3.5 cells, control cells and test cells were added and cultured for 24 h. As shown in Fig. 1b, c, the protein extract of HepG2 cells stimulated the test cells to produce much more IL-2 than was the case for control cells, HMBPP could not stimulate the test cells and control cells to produce IL-2, suggesting that J.RT3-T3.5-transfected cells expressing HCC-specific $\gamma\delta$ TCRs were successfully generated.

Selection of three candidate peptides from a PDL using artificial $\gamma\delta$ T cells

To identify epitopes recognized by HCC-specific transfected cells, we performed $\gamma\delta$ TCR-mediated biopanning of a 12-mer random-peptide PDL. The screening method was based on our previous paper [15]. The peptides by screening are listed in Table 2. We selected 3 prominent peptides as $\gamma\delta$ TCR-recognized peptide candidates based on their high frequency of appearance, designating them as HCC-specific peptide (HP). Next, HP1, HP2 and HP3 were chemically synthesized for further analysis.

BLAST analysis

We performed a protein BLAST search to identify related proteins containing the sequences for HP1, HP2 and HP3. The results are listed in Table 3 according to the E value score. Among these, the most matched protein from Homo sapiens with peptide HP1 is Hepatocyte growth factor-like protein.

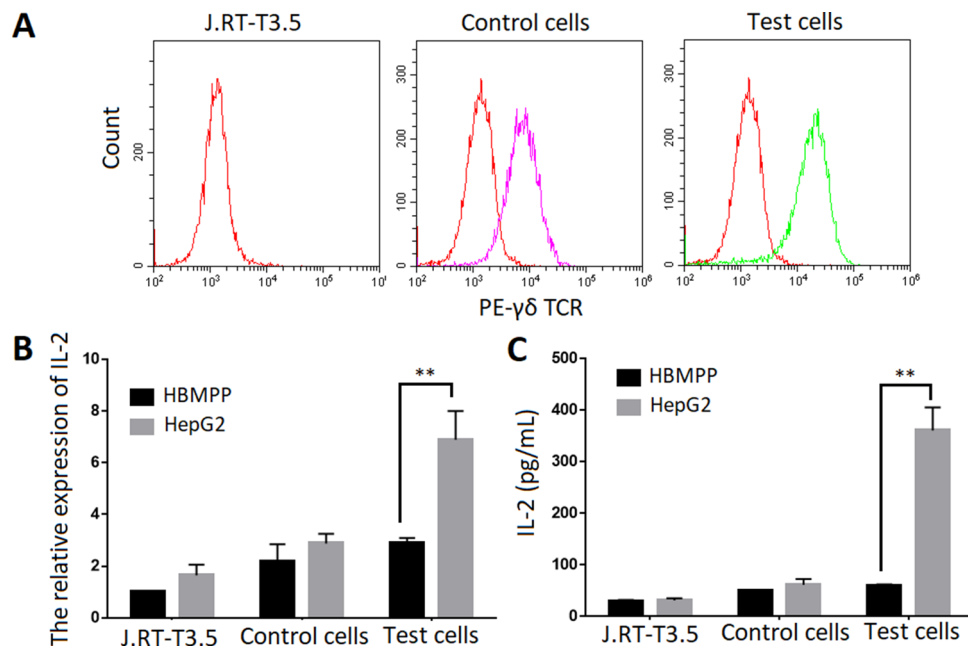


Fig. 1 Construction of transfected cells expressing HCC-specific and non-specific $\gamma\delta$ TCRs. **a** Assessment of $\gamma\delta$ TCRs expressed in J.RT3-T3.5-transfected cells via flow cytometer analysis. Based on predominant $\gamma 9$ and $\delta 2$ CDR3 sequences in HCC listed and highlighted in Tables S1 and S2 we developed two artificial cell lines expressing HCC-specific $\gamma 9/\delta 2$ TCR-transfected cells (test cells) and healthy control-transfected cells (control cells). The full-length $\gamma 9$ and $\delta 2$ chains were cloned into pREP7 and pREP9 expression vectors and co-transfected into J.RT3-T3.5 cells by electroporation. After 4 weeks of selection using G418 and hygromycin, Cells were stained with PE- $\gamma\delta$ TCR monoclonal antibody. The positive cells were further enriched by MACS and exceeded 90%. **b** The protein extract of HepG2 cells stimulated the test cells to produce much more IL-2 than

was the case for control cells by real-time PCR. Soluble phosphoantigens, HMBPP (1 ng/mL) and protein extracts of HepG2 cells were pre-coated with 96-well plate. J.RT3-T3.5 cells, control cells and test cells were added and cultured for 24 h. IL-2 secretion of control and test cells was measured by RT-PCR. **c** The protein extract of HepG2 cells stimulated the test cells to produce much more IL-2 than was the case for control cells by ELISA. Soluble phosphoantigens, HMBPP (1 ng/mL) and protein extracts of HepG2 cells were pre-coated with 96-well plate. J.RT3-T3.5 cells, control cells and test cells were added and cultured for 24 h. The IL-2 production in the supernatant of cell culture medium was measured by ELISA. **Denote $P < 0.01$. Data are shown as the mean of three independent experiments

Table 2 The amino acid sequences of 12 peptides and multiple sequence alignment

Name	Sequence	Frequency
HP1	TSRHQSWSPQDL	13/40
HP2	YQLLTPTSETFSY	9/40
HP3	TPLYKELDLPLQ	8/40
HP4	ATEVAKSESLH	2/40
HP5	YKILPHGQWRRS	2/40
HP6	TDAPESDVDRD	2/40
HP7	KHPVYPFDPSRP	2/40
HP8	AQTPVSYSPTTF	2/40

Confirmation of the peptide's capability to bind to HCC-specific transfected cells

Using the Heliquest website, we predicted the spiral structure of HP1, HP2 and HP3; this structure is presented in Fig. 2a. We performed binding assays to determine whether

the synthesized peptides could specifically bind to test cells. After we co-cultured the peptides (10 μ g) with such cells, FACS results showed that HP1 was better able to bind to test cells than that of control cells (Green line represents HP1; Fig. 2b). Next, we confirmed that HP1 could also stimulate test cells to secrete IL-2 in a dose-dependent manner (Fig. 2c), while control cells have no this ability. Collectively, these data suggested that HP1 could specifically bind to HCC-transfected cells.

HP1 could bind to and activate $\gamma\delta$ T cells from HCC patients in vitro

To investigate whether natural $\gamma\delta$ T cells could recognize and become activated by the HP1 peptide, we detected HP1's ability to bind to $\gamma\delta$ T cells isolated from HCC patients and healthy controls, as well as its ability to induce natural $\gamma\delta$ T cells proliferation. After $\gamma\delta$ T cells from 10 HCC patients and 10 healthy subjects were sorted by MACS, control peptide and HP1 peptide were co-cultured with these $\gamma\delta$ T cells

Table 3 Blast analysis of candidate peptide (HP1, HP2 and HP3)

No	Reference	Protein	Species	E value	Matching part
HP1	XP_011532034.1	Hepatocyte growth factor-like protein isoform X1	Homo sapiens	34.1	-TSRHQSW-PQ-
HP1	MCB09754.1	Immunoglobulin heavy chain junction region	Homo sapiens	25.7	-RHQ-WSP-D-
HP1	NP_001315537.1	Pentraxin-4 isoform 1 precursor	Homo sapiens	25.2	-R-Q-WSPQ-
HP2	AHA56119.1	Immunoglobulin heavy chain variable region	Homo sapiens	25.7	-LLTPSET-S-
HP2	NP_001093634.1	Phosphatase and actin regulator 2 isoform 1	Homo sapiens	24.8	-QLLTP-SE-FS-
HP2	AAF61928.1	Protocadherin fat 2	Homo sapiens	23.1	Y-LL-P-E-FS
HP3	XP_024309834.1	farnesyl pyrophosphate synthase isoform X1	Homo sapiens	26.1	-LY-ELDLP-LQ-
HP3	2RAH_A	Chain A, human Fdps synthase in complex with novel inhibitor	Homo sapiens	26.1	-LY-ELDLP-LQ-
HP3	3CP6_A	Chain A, farnesyl pyrophosphate synthetase	Homo sapiens	26.1	-LY-ELDLP-LQ-

under the condition with or without functional blocking antibody for 30 min. FACS results showed that $\gamma\delta$ T cells from HCC patients with HP1 had greater binding ability than such $\gamma\delta$ T cells from healthy controls ($P < 0.01$; Fig. 3a). Meanwhile, the binding of HP1 with $\gamma\delta$ T cells from HCC patients could be blocked by functional blocking antibody. It means that HP1 could specifically bind to $\gamma\delta$ T cells from HCC patients in vitro. Moreover, HP1 was shown to be able than the control peptide to induce proliferation of $\gamma\delta$ T cells in PBMCs from the same HCC patients ($P < 0.05$; Fig. 3b). 74.16 percent of these $\gamma\delta$ T cells belong to $\gamma 9/\delta 2$ T cells (Figure. S1A). Furthermore, to check the immune function of HP1-activated $\gamma\delta$ T cells, we detected cytokine production and cytotoxicity effect. The results showed that production of HP1-induced interferon gamma (IFN- γ) in $\gamma\delta$ T cells from HCC patients was higher than in $\gamma\delta$ T cells from healthy controls ($P < 0.05$; Fig. 3c). Levels of other cytokines such as tumor necrosis factor-alpha (TNF- α), IL-4 and IL-17 did not differ between $\gamma\delta$ T cells from HCC patients and those from healthy controls. In addition, we found that HP1-activated $\gamma\delta$ T cells could kill HepG2 cells effectively ($P < 0.01$; Fig. 3d).

Effector function of MSP-induced $\gamma\delta$ T cells from HCC patients

Using the OncoLnc website to demonstrate the relationship between MSP and survival probability for HCC patients, we found that patients with higher MSP expression had longer survival probabilities (Fig. 4a). Then, to investigate whether MSP could activate natural $\gamma\delta$ T cells, we detected the percentage of $\gamma\delta$ T cells induced by MSP. The results showed that MSP was better able to induce proliferation of $\gamma\delta$ T cells in PBMCs from the same HCC patients than in PBMCs from control protein ($P < 0.01$; Fig. 4b). 42.35 percent of these $\gamma\delta$ T cells belong to $\gamma 9/\delta 2$ T cells (Figure. S1B). To check the immune function of MSP-activated $\gamma\delta$ T cells, we detected cytokine production and cytotoxicity effect, and we found

that production of IFN- γ ($P < 0.01$) in MSP-induced $\gamma\delta$ T cells from HCC patients was higher than in such cells from healthy controls (Fig. 4c). We found that functional blocking antibody for $\gamma\delta$ TCR caused a reduction in IFN- γ production in MSP-stimulated $\gamma\delta$ T cells (Figure S2A). It was suggested that the activation of $\gamma\delta$ T cells by MSP is depend on $\gamma\delta$ TCR. Furthermore, we observed that MSP protein-activated $\gamma\delta$ T cells could kill HepG2 cell lines ($P < 0.05$; Fig. 4d). Collectively, these results suggested that $\gamma\delta$ T cells activated by MSP might act against HCC by secreting cytokines and/or by exerting an CTL-like effect.

Discussion

Selection of target antigens is a key determinant of the specificity and effectiveness of the CAR-T immunotherapy method [18]. In recent years, the application of CAR-T therapy based on $\gamma\delta$ T cells in tumor immunotherapy has attracted more and more attention. However, tumor antigens recognized by $\gamma\delta$ T cells are rarely identified, which has become the main restriction on this application of $\gamma\delta$ T cells. In this study, we used previously established methods to identify one candidate protein and one candidate peptide recognized by $\gamma\delta$ T cells [15]. J.RT-T3.5 cells do not express any TCR. When the full-length $\gamma 9$ and $\delta 2$ sequences are transfected into the cells, $\gamma\delta$ TCR is expressed on their surface. The cells secrete IL-2 after being stimulated by corresponding antigens. This method has also been used to identify mycobacterium TB protein antigens recognized by $\gamma\delta$ T cells [17]. In present study, we first investigated the diversity of the $\gamma 9/\delta 2$ T-cell immunorepertoire by sequence analyses of the expressed CDR3 in HCC patients. The difference of $\delta 2$ TCR chain was most pronounced presumably because of the increased potential for diverse sequences due to the additional D gene segment rearrangements and nucleotide additions/substitution [19]. Here, a common $\delta 2$ CDR3 sequence was found in PBMC of HCC patient's

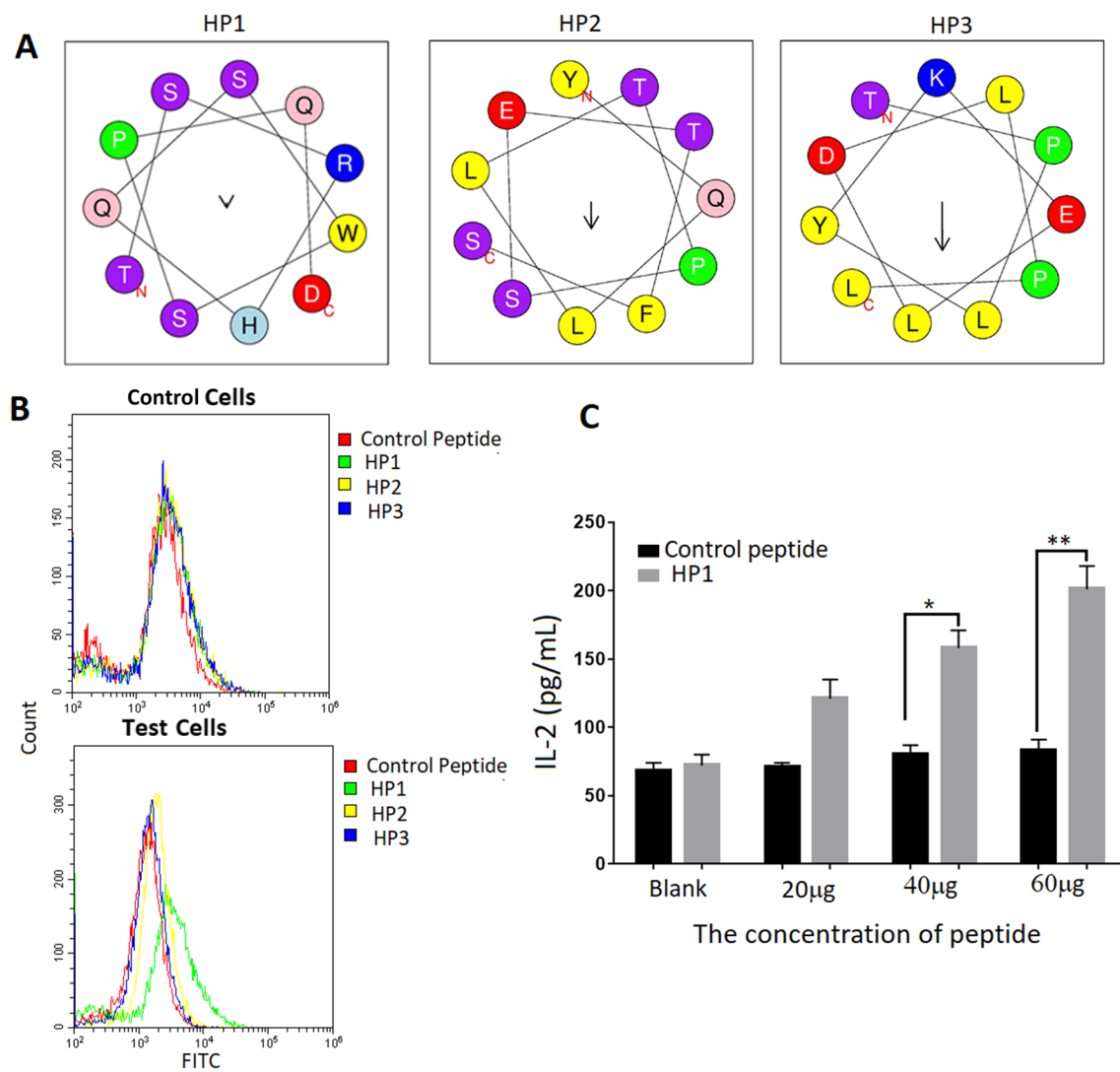


Fig. 2 Confirmation of the peptide's capability to bind to HCC-specific transfected cells. **a** The spiral structure of three peptides were predicted by the Heliquist website http://heliquist.ipmc.cnrs.fr/?tdsourcetag=s_pcqq_aiomsg. Enter the sequence of HP1, HP2 and HP3 in the blank. Helix type is 3–10, and click the button of process. The predicted spiral structure of the three peptides was presented. **b** HP1 was able to bind to test cells by FACS analysis. Three peptide and control peptide conjugated with FITC (10 µg) were co-cultured with test cells and control cells,

respectively. The results showed that HP1 was better able to bind to test cells than that of the control cells (Green line represents HP1). Results are representative of three independent experiments. **c** HP1 could also stimulate test cells to secrete IL-2 in a dose-dependent manner. Different dosages of HP1 and control peptide were co-cultured with test cells and control cells for 24 h. Production of IL-2 in the supernatant of cell culture medium was detected by ELISA. Data are shown as the mean of three independent experiments. *Denote $P < 0.05$. **Denote $P < 0.01$

groups. Our sequence analysis revealed that most Vδ2 T cell isolated from HCC patients and healthy controls also carried a hydrophobic amino acid residue (leucine/valine/arginine) at conserved position 97 (Table S2). The results suggested that the hydrophobic amino acid residue is not a prerequisite for γδ T cell reactive to HCC patients.

Then, we constructed γ9/δ2 TCR-transfected cells expressing the predominant HCC CDR3 sequence, identified a group of peptides capable of binding specifically to γδ T cells and candidate peptide-matched proteins by

bioinformatics analysis. Thus, we obtained a candidate HCC protein and peptides recognized by γδ T cells. They were able to bind to γδ TCRs with the HCC CDR3 sequence on the transfected cells and stimulate the expansion of γδ T cells from HCC patients.

New antigen recognized by γδ T cells that we identified is macrophage-stimulating protein (MSP), also known as hepatocyte growth factor-like protein. It is a kind of glycoprotein synthesized in the liver, and it has immunoregulatory activity [20]. MSP may bind to its specific

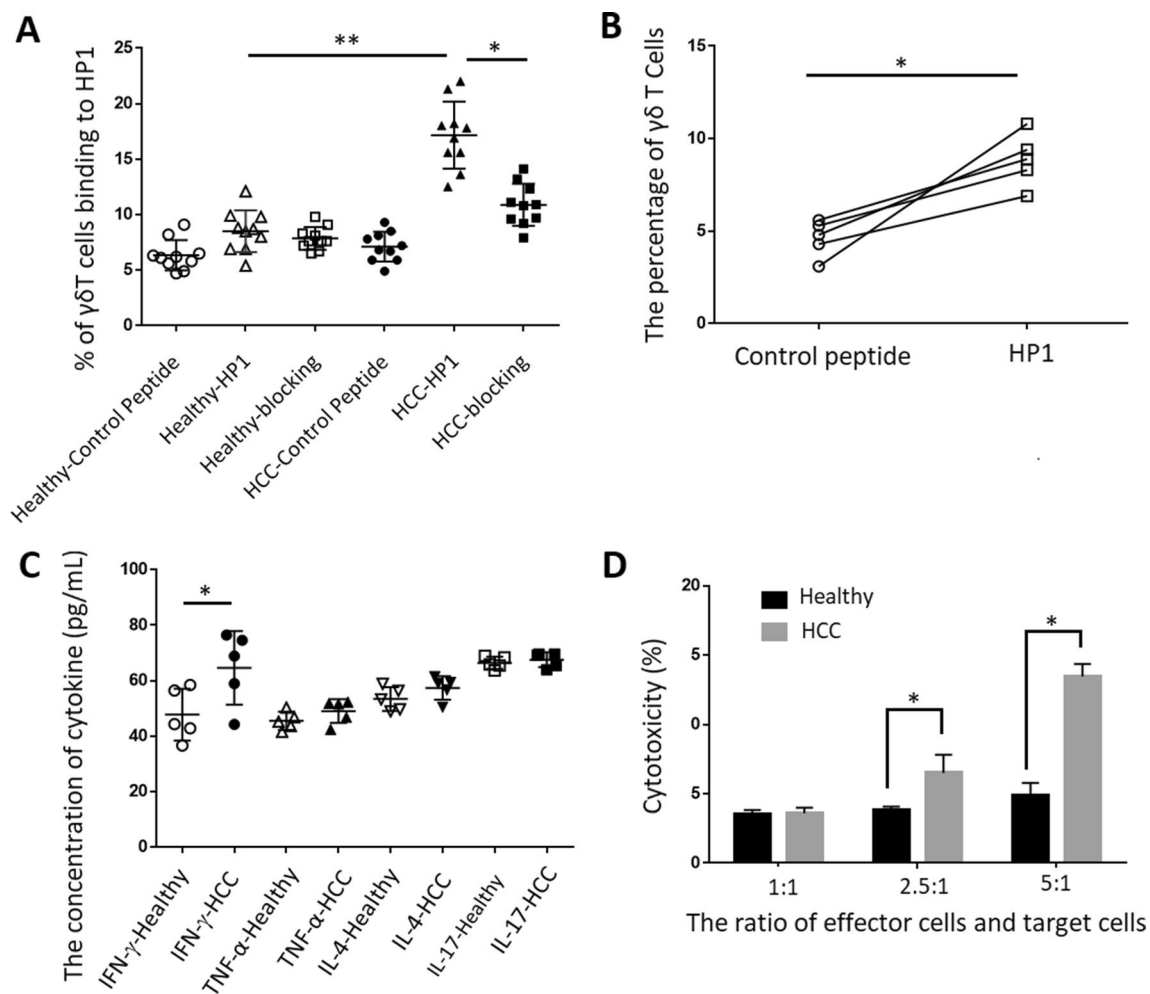


Fig. 3 HP1 could bind to and activate $\gamma\delta$ T cells from HCC patients in vitro. **a** HP1 bound to more $\gamma\delta$ T cells from HCC patients than $\gamma\delta$ T cells from healthy controls. $\gamma\delta$ T cells in PBMCs from 10 HCC patients and 10 healthy controls were obtained by MACS. HP1 and control peptide conjugated with FITC were co-cultured with cells under the condition with or without functional blocking antibody (10 ng/ μ L) for 30 min at 4 °C. The binding of HP1 and control peptide to $\gamma\delta$ T cells was measured by FACS. **b** Immobilized HP1 could induce proliferation of $\gamma\delta$ T cells from HCC patients. Immobilized HP1 and control peptide were pre-coated onto microtiter plates, on which $\gamma\delta$ T cells from 5 HCC patients were incubated for 2 weeks.

The percentage of $\gamma\delta$ T cells was measured by FACS. **c** HP1-activated T cells could secrete Th₁-based cytokines. Immobilized HP1 was pre-coated on microtiter plates, on which $\gamma\delta$ T cells from 5 HCC patients and 5 healthy controls were added. Supernatants were collected after 48 h and measured for IFN- γ , TNF- α , IL-4 and IL-17 secretion by ELISA. **d** HP1-induced $\gamma\delta$ T cells from HCC patients have the cytotoxicity of to HepG2 by CytoTox 96 non-radioactive cytotoxicity assay. After $\gamma\delta$ T cells from HCC patients inoculated by HP1 were sorted and cultured, cytotoxic activities of $\gamma\delta$ T cells against HepG2 in vitro was evaluated. Data are shown as the mean of three independent experiments. *Denote $P < 0.05$. **Denote $P < 0.01$

Ron-receptor tyrosine kinase to stimulate macrophages to regulate macrophage motility and its own phagocytic activity and to induce the production of oxygen-free radicals and cytokines; this in turn induces activation of nuclear factor kappa-light-chain-enhancer of activated B cells (NF- κ B) and regulates cell differentiation, migration and matrix invasion [21]. Additionally, MSP could reduce palmitic acid (PA)-induced lipid accumulation and lipogenesis in the HepG2 cell line [22]. Satoru et al. demonstrated that plasma MSP levels are associated with prostate cancer progression [23]. Our bioinformatics analysis

showed that HCC patients with higher expression of MSP had longer survival probability (Fig. 4a), suggesting the potential significance MSP in the diagnosis and progression of HCC.

MSP consists of six domains: an N-terminal domain homologous to plasminogen preactivation peptide; four copies of the kringle domain; and a C-terminal serine proteinase homology domain [24]. SWISS-MODEL (<http://swissmodel.expasy.org/>) were performed to predict the structure of MSP using homology modeling techniques. The result was presented in Figure S2B. However, due to the diversity

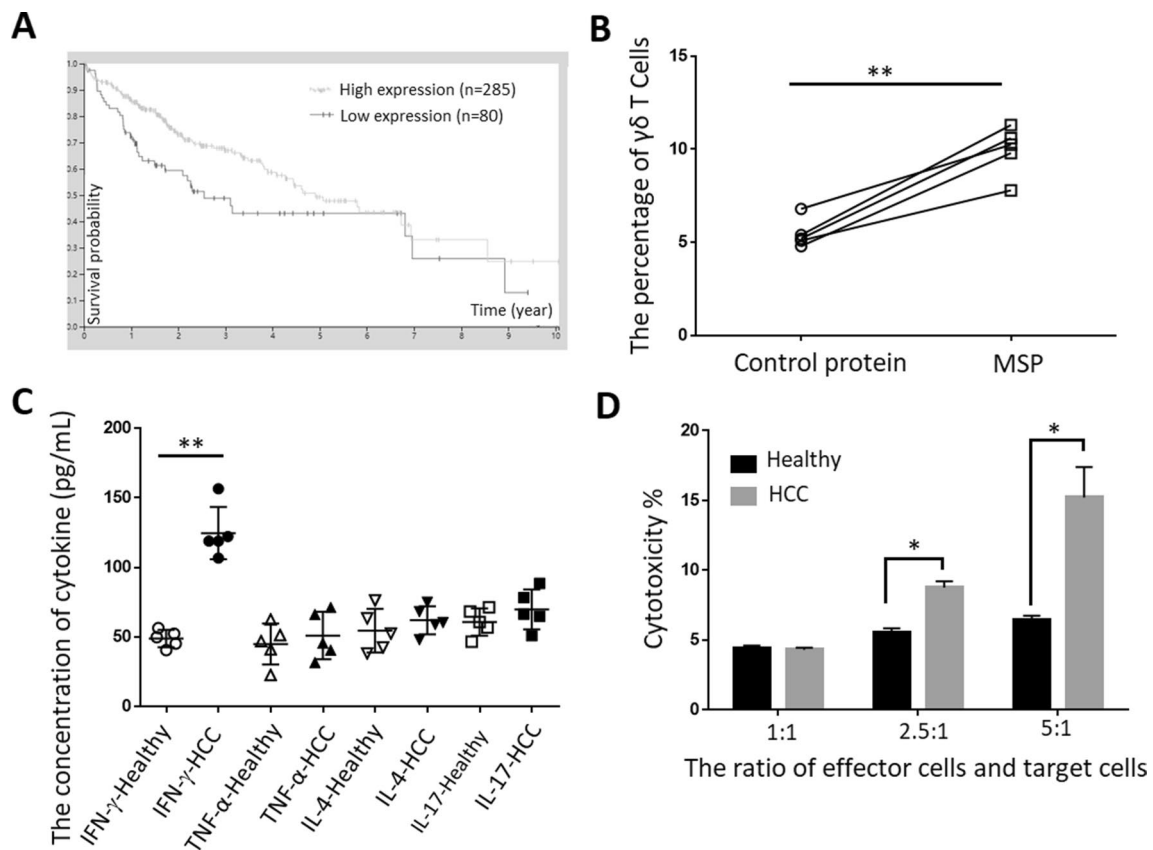


Fig. 4 Effector function of MSP-induced $\gamma\delta$ T cells from HCC patients. **a** The relationship of MSP with the survival probability of HCC patients, as shown by the OncoLnc website. In the oncolnc website, MSP was entered to get the Cox regression results. HCC is selected, lower percentile was set on 10 and higher percentile was set on 90, the Kaplan plot for MSP in HCC was determined. **b** Immobilized MSP could induce proliferation of $\gamma\delta$ T cells from HCC patients. Immobilized MSP and control protein were pre-coated onto microtiter plates, on which $\gamma\delta$ T cells from 5 HCC patients were incubated for 2 weeks. The percentage of $\gamma\delta$ T cells was measured by FACS. **c**

MSP-activated T cells could secrete Th_1 -based cytokines. Immobilized MSP was pre-coated on microtiter plates, on which $\gamma\delta$ T cells from 5 HCC patients and 5 healthy controls were added. Supernatants were collected after 48 h and measured for IFN- γ , TNF- α , IL-4 and IL-17 secretion. **d** MSP-induced $\gamma\delta$ T cells from HCC patients have the cytotoxicity of to HepG2 by CytoTox 96 Non-radioactive Cytotoxicity Assay. After $\gamma\delta$ T cells from HCC patients inoculated by MSP were sorted and cultured, cytotoxic activities of $\gamma\delta$ T cells against HepG2 in vitro was evaluated. Data are shown as the mean of three independent experiments. *Denote $P < 0.05$. **Denote $P < 0.01$

of $\gamma\delta$ TCR, we cannot predict the binding of $\gamma\delta$ TCR and MSP by molecular docking.

$\gamma\delta$ T cells recognize a wide variety of antigens, such as lipids, proteins, and phosphoantigens (P-Ag). The crystallographic structure of $\gamma\delta$ TCRs shows that CDR3 δ might be in direct contact with peptides and MHC-like protein and serve as a key determinant of specificity of antigen recognition [25]. He et al. demonstrated that recognition of $\gamma\delta$ T cells by human mutS homolog 2 (hMSH2) depends on $\gamma\delta$ TCRs and another activating receptor, NKG2D [26]. Other activating receptors, such as toll-like receptors (TLRs), also may provide costimulatory signals for $\gamma\delta$ T cell activation by different non-MHC protein ligands [27]. The mechanism by which $\gamma\delta$ T cells recognize MSP protein also needs further study.

We further evaluated the function of these $\gamma\delta$ T cells activated by candidate protein and peptides. They performed their major immunological effector functions in response to

HCC mainly via secreting cytokines and performing cytotoxic effect. Based on two mechanisms, we detected the roles played by MSP protein and HP1 peptide-activated $\gamma\delta$ T cells against HCC.

Cytokines secreted by $\gamma\delta$ T cells, like those secreted by $\alpha\beta$ T cells, include Th_1 -, Th_2 - and Th_{17} -type cytokines. $\gamma\delta$ T cells are thus divided into IFN- γ^+ $\gamma\delta$ T cells, IL-4 $^+$ $\gamma\delta$ T cells and IL-17 $^+$ $\gamma\delta$ T cells [28]. HCC is a typical inflammation-related cancer; various cytokines play important roles in its tumorigenesis and development. IFN- γ and TNF- α are Th_1 -type, IL-4 and IL-10 are Th_2 -type and IL-17 is a Th_{17} -type cytokine, respectively. As a key cytokine in the control of HCC, IFN- γ can activate effector cells and cytotoxic T cells, which can kill or inhibit HCC [29, 30]. TNF- α is an important inflammatory factor and has been confirmed to promote tumor growth and poor prognosis in HCC [31, 32]. IL-4 is produced by Th_2 cells, mast

cells, basophils and natural-killer (NK) cells. Like other cytokines, it can have a variable effect on a number of different target cells, including B cells, T cells, monocytes, endothelial cells and hepatocytes [33, 34]. IL-4 plays important roles in regulating antibody production, in the development of effector T-cell responses, in hematopoiesis and in inflammation. In the inflammatory state, IL-4 can inhibit proinflammatory cytokines and chemokine production in monocytes, and it enhances the expression of vascular cell adhesion molecule 1 (VCAM-1) in endothelial cells. IL-10 is a pleiotropic cytokine produced by immune cells and has been shown to inhibit various immune reactions (35). In HCC, IL-17⁺ T cells have been found in increased numbers within tumors and correlate with poor survival rates and increased postoperative recurrence, indicating that Th₁₇ cells and IL-17 may promote tumor progression in HCC [36]. In our study, we first detected cytokines secreted by $\gamma\delta$ T cells that had been activated by MSP protein and HP1. As shown in Fig. 3c, the production of IFN- γ in HP1-induced $\gamma\delta$ T cells from HCC patients was higher than in such cells from healthy controls; so was the production of IFN- γ in MSP-induced $\gamma\delta$ T cells from HCC patients (Fig. 4c). Levels of other cytokines, such as TNF- α , IL-4 and IL-17, did not differ between $\gamma\delta$ T cells activated by MSP protein and HP1 from HCC patients and those from healthy controls. Thus, MSP protein cells and HP1 peptide stimulated V γ 9V δ 2 T showed potential as Th₁ biased immune reaction for HCC immunotherapy.

We then determined the cytotoxicity of $\gamma\delta$ T cells activated by MSP protein and HP1 to the HCC cell line HepG2. $\gamma\delta$ T cells expanded by MSP protein and HP1 had cytotoxic effects on HepG2 cells (Figs. 3d, 4d). However, we verified this function of $\gamma\delta$ T in only 10 HCC patients in the current study. We intend to expand the sample size in subsequent research.

In conclusion, our findings add to the understanding of the mechanism by which $\gamma\delta$ TCRs recognize protein antigens. We identified a peptide (HP1) and a protein (MSP) as new candidate HCC antigens recognized by $\gamma\delta$ T cells. They could be potential candidate peptides or antigenic components to develop new targets for immunotherapy, as well as biomarkers of HCC. Further studies are in progress to confirm the bio-function of MSP protein and the HP1 peptide in $\gamma\delta$ T cell-based anti-HCC therapy.

Author contributions Conceived and designed the experiments: XX, BD. Performed the experiments: MZ, YG, FQ. Analyzed the data: XX BD. Contributed reagents/materials/analysis tools: GL FL. Wrote the paper: XX BD.

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Compliance with ethical standards

Conflict of interest The authors declare no conflicts of interest.

Ethical approval This work received approval from the Clinical Ethics Committee of Hubei University of Medicine (2019-TH-024).

Informed consent All individuals gave their informed consent to participate.

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